Copenhagen Business School

Project

Machine Learning and Deep Learning

Automated Image-Based Waste Classification Using Machine Learning Models

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Abstract

As global waste management continues to be a growing issue, the automation of waste classification presents a valuable opportunity to enhance recycling. In this project, we explore how machine learning models can improve waste classification through automated image-based recognition and address the question "How can recycling systems be improved by applying machine learning classification?" Our goal is to build such models, both traditional and deep learning-based, that can effectively classify images of waste across 30 distinct categories. We implement and evaluate three primary approaches: a traditional Support Vector Machine (SVM) model with Principal Component Analysis (PCA) for dimensionality reduction, which serves as a baseline offering insights into the capabilities of conventional machine learning. We then developed two progressively more complex custom-built Convolutional Neural Networks (CNNs), including an enhanced model with residual connections to facilitate deeper feature extraction and finally we fine-tuned a pre-trained EfficientNetB0 model. We trained and evaluated these models using metrics such as accuracy, precision, recall, and F1-score on a balanced dataset of 15,000 images, which contains 30 different waste categories, including both studio-quality and real-world images.

Our results demonstrate a clear performance progression across model types. The SVM performed reasonably well with an accuracy of 55% and an F1 score of 0.5600, the deep learning models significantly outperformed it, with the EfficientNetB0 achieving the highest test accuracy of 87.39% and an F1 score of 0.8734. Our findings highlight that while conventional machine learning models can perform reasonably well in image-based waste classification tasks, CNNs have emerged as a dominant choice for a reason. Their ability to automatically learn hierarchical spatial features from images makes them well suitable for visual classification problems. However, handling a large number of waste categories is challenging and requires models that are capable of distinguishing small differences and extracting deeper features. Advanced deep learning architectures are outstanding in this context, especially models that leverage transfer learning, such as EfficientNetB0. By leveraging features learned from large-scale datasets, they converge faster, generalize better, and achieve higher accuracy, which makes them powerful and also practical for real-world applications. We therefore recommend transfer learning-based CNNs for future implementations of waste classification systems, especially in applications requiring high accuracy across diverse categories.

Keyword: Automated waste recycling, Convolutional Neural Network, Support Vector Machine, Image Classification, Transfer learning

1. Introduction

Effective waste management is a growing concern worldwide, as increasing urbanisation and growing consumption lead to more waste. One of the key challenges in this respect is the accurate classification of waste materials, which is crucial to automate recycling processes and promote environmental sustainability. Traditional methods of waste sorting usually rely on manual sorting, which is labour-intensive, time-consuming and prone to human error. To assess these limitations, our project investigates the application of machine learning techniques to automate waste classification.

Our aim is to address the question "How can recycling systems be improved by applying machine learning classification?" To answer this question, we focus on developing and evaluating conventional and deep machine learning models that can accurately distinguish between predefined waste categories based on visual features. We aim to train algorithms that can not only efficiently sort the waste and hereby contribute to smarter waste management systems, but also scalable and able to handle a wide range of waste categories. This is crucial for real-life applications, where the diversity of waste materials often necessitates the classification of many different categories. With this goal in mind, the models must demonstrate high accuracy and robustness across a wide variety of waste types, ensuring their viability in practical recycling environments.

In pursuit of the main research question, the following sub-questions are addressed:

- What types of machine learning models are most effective for image-based waste classification?
- How does the performance of traditional models compare to deep learning approaches?
- Can models that utilize transfer learning further enhance classification accuracy?

We aim to answer these research questions by building and evaluating four different machine learning models, including convolutional and deep learning models on 15,000 images. We will compare and assess the models performance with particular attention to their robustness and ability to generalize across diverse waste categories. We aim to provide insights into the feasibility, scalability and ability to adapt into real-world situations, and their potential contribution to more efficient and sustainable recycling practices.

2. Related Work

In the last few years several papers have been published on the topic of waste management using artificial intelligence and machine learning models. One of the most common approaches is using Convolutional Neural Networks (CNNs), which have proven to be very effective in classifying images.

Poudel and Poudyal (2022) leveraged CNN-based pre-trained models, such as InceptionResNetV2, VGG19, MobileNet and so on, on a waste dataset containing 7 classes. Their results show that InceptionResNetV2 achieved the best accuracy (93,82%) while VGG19 performed worst (71,81%). This paper demonstrated that various deep CNN models can achieve very different results. Basheer

Ahmed et al. (2023) also used deep CNNs for classifying waste. They compared a custom built CNN with transfer learning models, such as DenseNet169, MobileNetV2, and ResNet50V2 models using 5 waste categories. The baseline CNN achieved an accuracy of 88.5%, while the pre-trained models outperformed with 94.4% (DenseNet169), 97.6% (MobileNetV2) and 98.9% (ResNet50V2) accuracy. The results of this study show that modern, pre-trained models, which utilize transfer learning can deliver outstanding performance.

In addition to traditional CNNs, EfficientNet has gained popularity as a strong alternative in waste classification tasks due to its ability to deliver high accuracy with optimized computational efficiency. For example, Risfendra et al. (2024) used transfer learning with EfficientNetB0 to classify 6 types of waste using a dataset of 7,014 images. Their model achieved an accuracy of 91.94%, demonstrating the architecture's practical effectiveness. Santoso et al. (2024) proposed an innovative hybrid model by combining EfficientNet-based transfer learning with PCA for dimensionality reduction in dry waste classification. Their model achieved an outstanding 99.54% accuracy on a dataset of 2567 images with 8 classes using this combined approach.

On the other hand, some researchers have explored more traditional machine learning approaches as Jangsamsi (2023) did. The paper focuses on extracting features using Fourier descriptors (FD), histogram of oriented gradients (HOG) or local binary patterns (LBP), and feeding them into an SVM for classification, which proved that conventional machine learning models can also provide competitive performance if they are combined with robust feature engineering techniques. Hanbal et al. (2020) also evaluated the performance of four traditional machine learning models, Random Forests, Gaussian Naïve Bayes, Support Vector Machines, and Multilayer Perceptron in order to classify biodegradable and non-biodegradable waste. They used a dataset with 15,000 images, where the SVM model achieved an accuracy of 89.51%. Puspaningrum et al. (2020) created a waste classification pipeline that utilized Scale-Invariant Feature Transform (SIFT) for feature extraction, followed by PCA for dimensionality reduction, and SVM for classification, which also demonstrated the effectiveness of conventional techniques for image classification tasks.

Based on the literature review conducted we concluded that SVM is one of the best suitable conventional models for classifying images. Among deep learning models, custom CNNs and transfer learning-based models such as EfficientNet have consistently demonstrated high performance in image classification tasks. Additionally, we observed that most of the reviewed studies focus on using datasets that are featuring a small number of unique classes, ranging between 4-7 waste categories. This can limit the generalization of these models to more complex or real world representations, where there is a much more diverse variety of classes. Classifying a larger number of categories still remains a challenging task. By using a bigger and more diverse dataset, and comparing traditional and modern deep learning approaches our goal is to present a more scalable and practical solution to improve classification performance in waste recycling.

3. Conceptual Framework

3.1 Data Preparation

The data preparation comprises four sequential steps — data combination & quality checking, data splitting, data normalisation, and data augmentation — summarised in Figure 1. The dataset used in this project includes a variety of waste items across different categories. Every category contains the waste item images of studio-like quality, and the images captured in real-world scenarios. These images are stored separately in two sub-folders of the main category folder.

To begin with the data preparation, all images of the category folder are collected together, and then the inspection of the image size, image quality, and image quantity per category is conducted before performing data splitting, which creates three sets of data — training data, validation data, and test data. Next, all data is normalised for the use of all models. Lastly, data augmentation is applied to certain models in order to increase the generalisation ability of the models.

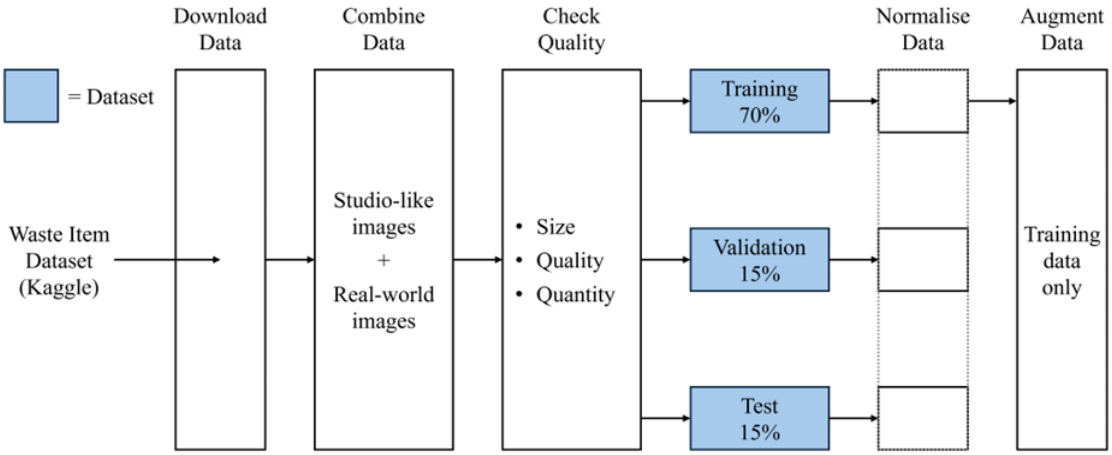


Figure 1

3.2 Training Models

This project includes four models: a baseline Support Vector Machine, two Convolutional Neural Networks, which is a widely used technique for image classification tasks, and lastly a customised EfficientNet Model, which is mainly used for model performance comparison.

All models are trained by using the data sourced from the original dataset; however, data splitting step is conducted for the SVM model and the CNN models separately. See Figure 2. Besides, data augmentation is not applied to the SVM model, as it relies on flattened and dimension-reduced features that are not image-based.

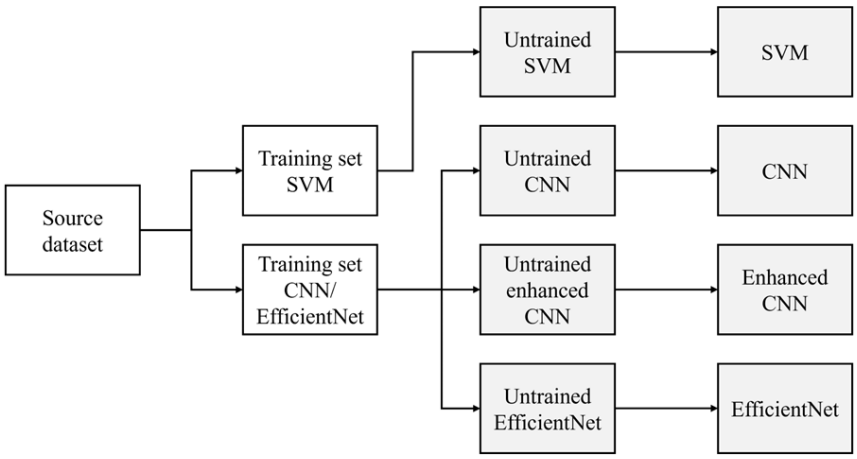


Figure 2

4. Methodology

4.1 Dataset Description

The dataset used in this project has been well-organised and contains various images from recyclable materials, general waste, and household items, including plastic, paper & organic waste, and so on. The images are classified into 30 distinct categories. See table 1. Every category folder comprises two sub-folders, named 'default' and 'real_world', each of which stores 250 images, making up 500 images per category and total 15,000 images in the dataset.

No.	Category	No.	Category
1	Plastic trash bags	16	Aluminum food cans
2	Plastic shopping bags	17	Styrofoam food containers
3	Plastic food containers	18	Styrofoam cups
4	Plastic soda bottles	19	Paper cups
5	Plastic cup lids	20	Cardboard boxes
6	Disposable plastic cutlery	21	Cardboard packaging
7	Plastic straws	22	Office paper
8	Plastic water bottles	23	Newspaper
9	Plastic detergent bottles	24	Magazines
10	Glass beverage bottles	25	Food waste
11	Glass food jars	26	Coffee grounds
12	Glass cosmetic containers	27	Tea bags
13	Aerosol cans	28	Eggshells
14	Steel food cans	29	Clothing
15	Aluminum soda cans	30	Shoes

Table 1

The ‘default’ sub-folder contains clean, high-quality images of waste items captured in standardised or studio-like environments. These images offer clear, unobstructed visual features of each item, making them highly suitable for training and evaluation of image classification models. In this project, these studio-quality images serve as the foundation for developing and validating the SVM model and the CNN models. Each image has a fixed resolution of 256×256 pixels, ensuring consistency across the dataset. See figure 3 for example images.



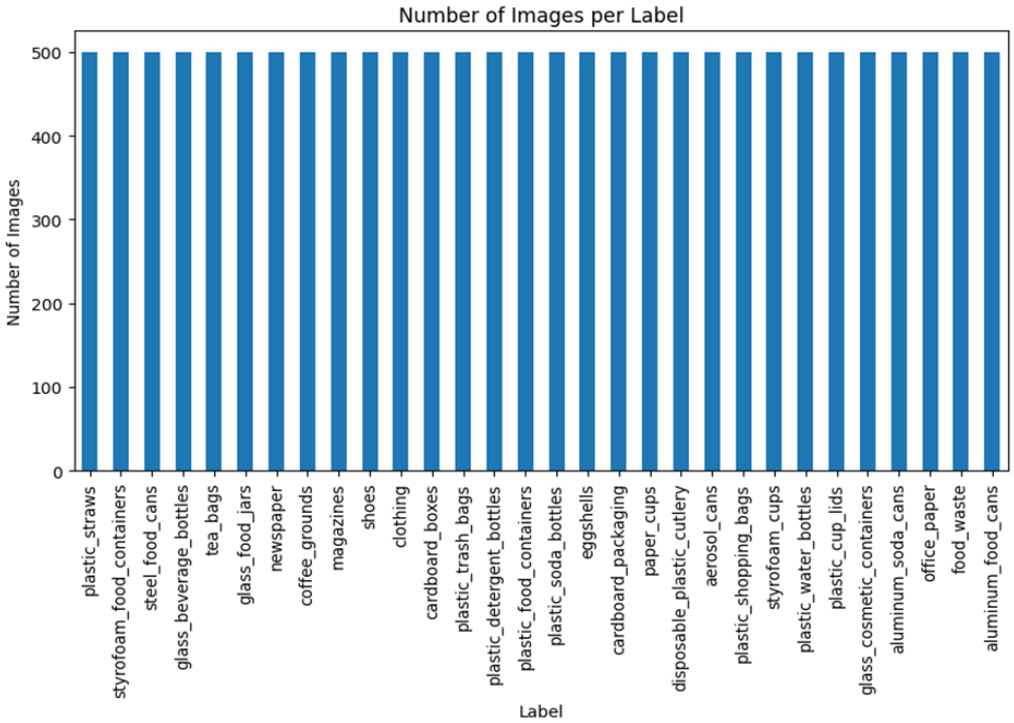
Figure 3

The ‘real_world’ sub-folder contains images of waste items captured in everyday, unstructured environments, such as public spaces, or cluttered backgrounds. These images reflect the variability and complexity of real-world scenarios, including changes in lighting, angle, and partial occlusion. When combined with the studio-like images from the ‘default’ sub-folder, these real-world samples increase the overall diversity of the dataset and expose the model to more realistic and challenging visual settings. See figure 4 for sample images.



Figure 4

Figure 5 shows the number of images per label in the dataset. Every waste category has exactly 500 images, therefore the classes are evenly balanced. This ensures equal learning opportunities for all categories, which is especially useful when dealing with multiple categories. It also helps make the performance results fairer and more consistent.



4.2 Data Pre-Processing

We apply distinct data preprocessing techniques for the SVM model and the neural network models (CNNs and EfficientNet) to optimize their performance. Data normalization is a fundamental step in ensuring consistency across datasets. For the SVM model, the images are resized from their original dimensions of 256×256 pixels to 32×32 pixels. This smaller size is sufficient for efficient training while retaining essential features for classification across 30 classes. Labels are converted from non-numerical labels to numerical values. Pixel values, initially ranging from [0, 255], are scaled to the [0, 1] range, which enhances training stability. Subsequently, the 32×32×3 images are flattened into 1D vectors of length 3,072, which are further standardized using StandardScaler() to achieve 0 mean and unit variance, a critical step due to the SVM model's sensitivity to feature scales.

For the neural network models, images are processed at their original size of 256×256 pixels. Labels are transformed into one-hot vectors. To perform a more robust training of the models, we choose to augment the training data. This involves creating modified versions of existing data to increase dataset diversity. We augment the data by randomly flipping images horizontally. Just like the SVM model, pixel values are rescaled to the [0, 1] range for the CNN model. However, for EfficientNet, the pixel values are maintained in their original range, as required by the model's architecture.

Additionally, we split the dataset into training (70%), validation (15%) and test sets (15%) for both SVM and neural network models. These preprocessing methods ensure that each model receives data in a format best suited to its design, thereby enhancing overall performance.

4.3 Data Analytics

4.3.1 Support Vector Machine

Support Vector Machine was developed as a baseline model. SVM is a machine learning algorithm that works well for classification tasks, especially in high dimensional spaces. In this case, the goal was to demonstrate the effectiveness of traditional machine learning models in the context of waste image classification, while also providing a benchmark for more complex models that would be evaluated later.

Previously, we performed several pre-processing steps. To further reduce the number of features while retaining the most important information, principal component analysis (PCA) was applied. We decided to keep 300 principal components to balance between computational efficiency and the preservation of important features . PCA helps reduce noise and computational complexity, which helps the SVM model perform better by focusing on the most significant features of the data.

For the SVM model, we used the Radial Basis Function (RBF) kernel, which is commonly used for non-linear classification tasks. The SVM classifier was integrated into a pipeline along with the PCA transformation. To improve the performance of the model, a GridSearchCV approach was used to tune the hyperparameters, in particular the regularization parameter (C) and the kernel coefficient (gamma). A range of values of these parameters was tested and the best combination was selected by using 5-fold cross-validation. The grid search tested values of C in [1, 5, 10, 50] and gamma in ['scale', 0.01, 0.001]. The optimal parameters determined were C=50 and Gamma='scale'. The default 'scale' setting for gamma automatically calculates it as the inverse of the product of the number of features and the variance of the input data:

$$\gamma = \frac{1}{n_{\text{features}} \cdot \text{Var}(X)}$$

This allows the model to choose a value for gamma based on the scale of the data, while maintaining a good balance between model complexity and generalization.

After determining the best hyperparameters, the SVM model was trained on the pre-processed data. We then evaluated the model performance on the test set and calculated the accuracy metrics. Additionally, a confusion matrix was generated to provide insight into the classification performance across the different waste categories.

4.3.2 Convolutional Neural Networks

Previously, we implemented an SVM model as a baseline model to evaluate the effectiveness and accuracy of basic machine learning in classifying waste images. To further compare the performance between different models, we proceeded to build two convolutional neural network (CNN) models. CNNs are widely used for image classification tasks and are well suited for handling visual data. The

aim is to examine whether customised deep learning models could outperform the SVM baseline model in terms of classification accuracy and generalisation.

Several preprocessing steps have been applied before, to prepare the dataset for training the CNN models. We built the first CNN model as a baseline to see how a simple deep learning approach would perform on the waste classification task. The model has two convolutional blocks. The first block includes a Conv2D layer with 32 filters, followed by a MaxPooling2D layer, while the second block uses 64 filters and applies max pooling as well. After the convolutional layers, the feature maps are flattened into a 1D vector and passed through a dense layer with 128 units. The final output layer has 30 units, matching the number of waste categories, and uses a softmax activation to produce class probabilities. This baseline model was intentionally kept simple so we could use it as a reference point when testing more advanced CNN architectures later in the project.

To train the first CNN model, we chose to use the Adam optimiser, which is a commonly used optimisation algorithm for deep learning tasks. Since this project is a multi-class classification task and we have one-hot encoded labels, we chose the categorical_crossentropy as the loss function. During the training, we also tracked the evaluation metrics to get a better picture of the model performance. It was trained for 10 epochs, with performance evaluated on the validation set after each epoch to monitor learning progress and detect potential overfitting.

After training the baseline model, we built a more advanced CNN that introduced residual connections to improve performance. In this model, residuals were added after every two convolutional blocks. The first block starts with a Conv2D layer with 32 filters, followed by ReLU activation and max pooling. The output from that block was then passed to another Conv2D layer with 64 filters and activation. To create the residual connection, the original pooled output was projected to match the number of filters and then added to the output of the second convolution. This pattern repeats in the third and fourth blocks, which use 128 filters. After the second residual connection, another max pooling layer was applied before flattening the feature maps and passing them through a dense layer with 128 units. A dropout layer with a rate of 0.5 was added to help reduce possible overfitting, and the final output layer uses softmax activation to produce class probabilities for the 30 waste categories as the first CNN model has.

The second CNN model with residual connections was compiled using the same settings as the first CNN model. The only difference in the training setup is the number of epochs, which we increased from 10 to 30 to give the model more time to learn. By building a deeper architecture and introducing residual connections, we hope to achieve better classification performance compared to the first simpler baseline CNN model.

4.3.3 EfficientNet-B0 Model

To obtain better results on the dataset, we trained another model based on the EfficientNet architecture. EfficientNet was introduced by researchers at Google Brain, represents a significant advancement in CNN design. Their work fundamentally addressed the challenge of scaling CNNs for better accuracy and efficiency (Tan & Le, 2019). EfficientNet models achieve great accuracy on image classification tasks while being more computationally efficient than previous architectures. The core method is compound scaling that uniformly scales network width, depth, and resolution with a fixed set of scaling coefficients. This balanced scaling approach allows EfficientNet to achieve better performance with fewer parameters compared to other contemporary models. This efficiency makes them particularly attractive for applications with limited computational resources (Tan & Le, 2019).

In our paper, the EfficientNet-B0 was selected as the base architecture for transfer learning. The model was initialized with weights pre-trained on the ImageNet dataset, which should be reusable in different domains. We will transfer this pre-trained EfficientNet-B0 model into our problem domain with fine-tuning. The EfficientNet-B0 model is set to be untrainable, but we unfroze the last 20 layers to allow the model to adjust its higher-level, more specialized features to the specifics of our waste dataset. We then add some new layers to turn the classification part into 30 categories.

The model was compiled using the Adam optimizer and categorical cross entropy as the loss function. Performance was monitored using accuracy, precision, recall, and F1-score. The training contained 15 epochs, including callbacks to optimize training duration and adjust the learning rate.

We chose this model because of the computational efficiencies that it provides, making it easier to train, and because it has proved to be an excellent model for multi-category image classification problems. We hypothesize that the EfficientNet-B0 model will perform better than the other models in this paper.

4.3.4 Evaluation Metrics

In our waste classification problem, we evaluated the performance of our model on the test dataset using four key metrics: accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of how well our model performs across different aspects of classification. The four metrics are calculated by the following:

$$Accuracy = \frac{Correct\ Predictions}{All\ Predictions}$$

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$$

$$F_1 = \frac{2}{Precision^{-1} + Recall^{-1}}$$

Accuracy is a straightforward measure of overall model performance. Precision is crucial when minimizing false positives is important, as it measures the model's ability to avoid incorrect positive predictions. Recall is useful when minimizing false negatives is critical, ensuring that all true positive cases are captured. Precision and recall are often in conflict, meaning that increasing one may cause a decrease in the other, which is why balancing them is a key challenge in classification problems. F1-score is calculated from precision and recall, providing a single value that balances the two. Therefore, maximizing the F1 means maximizing both the precision and recall.

By analyzing these four metrics together, especially the F1 score, we gain a detailed view of our models, enabling us to compare the performances.

5. Results

The test results show a clear performance difference across the four models. The best-performing model is the EfficientNet with fine-tuning, which achieved an F1 score of 0.873 on the test set. This was followed by the CNN with residual connections, which scored 0.661, and the baseline CNN, which reached 0.612. The SVM model, used as a baseline model in this project, got an F1 score of 0.560. The full details of precision, recall, and F1 score for each model are summarised in Table 2.

Model	Accuracy	Precision	Recall	F1 Score
SVM	0.5500	0.5800	0.5500	0.5600
Baseline CNN	0.6084	0.6731	0.5739	0.6116
CNN with Residual	0.6611	0.7342	0.6416	0.6606
EfficientNet	0.8739	0.8918	0.8642	0.8734

Table 2: Accuracy, Precision, recall, and F1 scores

The dataset includes 30 waste categories, and many of them are visually quite similar — for example, different kinds of plastic containers, paper products, or packaging materials. These similarities make it harder for the models to correctly distinguish between classes, especially for those without deep architectures or pretrained weights. This might be the reason why we have the lower accuracy and recall values seen in the simpler models. While the dataset is balanced in terms of the number of images per class, the overlap in visual characteristics across categories adds complexity to the task. This also explains why more complex models, like the CNN with residual connections, still made classification errors on the test set.

The baseline SVM model has moderate performance with an F_1 score of 0.560. It seems that it lacks the ability to learn deeper visual patterns in the data. The baseline CNN improved by achieving higher precision and recall, but it was still limited by its relatively simple architecture and a tendency to overfit.

The second CNN model, which included residual connections, performed better across all metrics. The improvement in recall and F_1 score suggests that the deeper architecture and skip connections helped the model learn more useful features and retain important information across layers. Finally, we tested the EfficientNet model with fine-tuning. As expected, it achieved the highest performance, reaching an F_1 score of 0.873. However, since it mainly relies on pretrained weights, it served as a benchmark to evaluate how well our customised models compared to a transfer learning approach.

For the baseline and residual CNN models, as well as the EfficientNet model, the training and validation loss curves can be found in Figure 5. For the baseline CNN, we can see that after around five epochs, the model starts to overfit as the validation loss begins to increase while the training loss continues to decrease. For the CNN with residual connections, both the training and validation loss curves show a more stable pattern. The validation loss does not drop as steeply but remains slightly fluctuated within a range and not an increasing trend, indicating improved generalisation. Looking at the EfficientNet model with fine-tuning, we can see a noticeably consistent decline in training loss, and the validation loss is considerably lower than all 3 other models. The EfficientNet model shows rapid improvement in the first few epochs and maintains a small gap between the two loss curves. Unlike the baseline and residual CNN models, the EfficientNet does not show signs of overfitting, and even though the validation loss slightly increases in the final epoch, it remains stable overall, indicating that the model has likely reached its optimal training point.

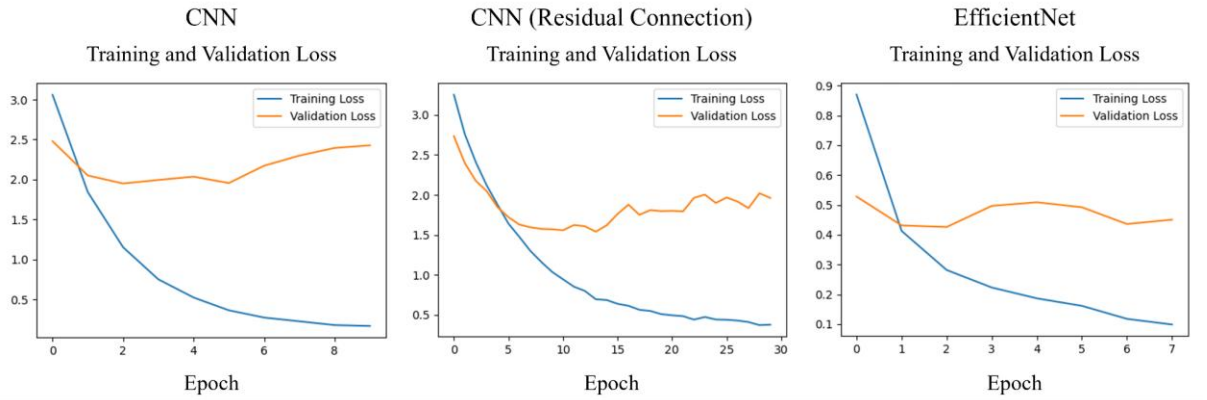


Figure 5: Training Loss Results

In addition to performance metrics, training time and model complexity are important factors to consider when evaluating different approaches. Among the four models, the SVM was the fastest, completing in 253.39 seconds, followed by the baseline CNN at 370.91 seconds, which trained for just 10 epochs. The EfficientNet model, despite using a pre-trained backbone and fine-tuning, completed training in 476.92 seconds, which is still quite efficient considering the model's size and complexity. The CNN with residual connections had the longest training time at 1,357.95 seconds, mainly because it was trained for 30 epochs — three times more than the baseline CNN, and also has more complex and deeper layers. Although it took longer to train, this model still delivered the best performance among the models built

from scratch. If we exclude EfficientNet and focus only on the models developed in this project, the residual CNN has the finest balance between improved accuracy and training time, considering the extended number of epochs. In practice, however, the EfficientNet model remains the most effective overall due to its pre-trained features and ability to reach the highest test accuracy.

6. Discussion

In the report, we investigated the potential of machine learning and deep learning models to improve waste classification systems. Our main research question, "How can recycling systems be improved by applying machine learning classification?" was addressed by evaluating four models in our project. The results demonstrate that the EfficientNet model significantly outperformed the other models. These findings suggest that transfer learning using pre-trained models, specifically EfficientNet, is highly effective for image-based waste classification that has many categories. Additionally, the better performance of the CNN with residual connections over the baseline CNN highlights the value of deeper architectures with skip connections for complex image classification tasks.

The result points out the importance of selecting appropriate machine learning architectures for complex multi-category waste classification. EfficientNet with pre-trained knowledge can achieve high performance to use in real world study. Moreover, the findings indicate that while traditional models like SVM can provide a starting point, their performance is limited when compared to more advanced neural networks. For future research, exploring more advanced models, such as EfficientNet-B7 or Vision Transformers (ViTs), may further enhance classification performance.

This study has some limitations. First, our model training was also constrained by computational resources, limiting the neural network training to a maximum of 30 epochs to reduce training time. With more powerful computing capabilities, it could potentially improve performance. Additionally, the classification task involved 30 waste categories, many of which have similar items. The high number of classes and the image similarity between certain categories increased the classification complexity and likely affected model performance.

The ethical considerations in this project primarily relate to the responsible use of data. Although the environment can benefit from improving recycling efficiency through machine learning, using images of waste can raise ethical concerns, particularly if they include identifiable brands, logos, or product packaging that could reveal user preferences or habits. Before deploying the model, people should consider that the dataset should not contain any identifiable information related to individuals.

7. Conclusion & Future Work

In this project, we explored the application of different machine learning techniques that can effectively classify waste images. Our goal was to evaluate the efficiency of the different models, comparing both traditional and deep learning methods. Starting with a traditional Support Vector Machine with PCA as

a baseline, followed by two progressively more complex CNN architectures, including an enhanced model with residual connections and culminating in a fine-tuned EfficientNetB0 model, we evaluated each approach on a comprehensive dataset containing 15,000 images across 30 categories.

The results clearly indicate that deep learning models are better suitable for this task. The SVM served as a useful baseline, providing insight into how traditional models perform when combined with basic dimensionality reduction techniques. While it offered reasonable accuracy (55%), it struggled to learn deeper visual patterns in the data. The introduction of convolutional architectures significantly improved classification performance. Our custom CNNs, designed to learn hierarchical feature representations directly from image data, showed great improvements, especially the residual variant (66%). The best-performing model, however, was the fine-tuned EfficientNetB0, which leveraged pretrained knowledge from large-scale datasets and adapted effectively to the waste classification task. It achieved a test accuracy of 87.39% and an F1 score of 0.8734, outperforming both the traditional and custom models. These results highlight the advantages of transfer learning and deeper feature extraction in complex visual classification tasks.

One of the biggest challenges of the project was to classify waste into 30 different categories. In such a large number of categories, there are many which are visually very similar, which introduces extra complexity. Our results show that while traditional machine learning models can be suitable for simple settings, deep learning models are better adapted to tackle these challenges and scale to real-world situations.

Despite the promising results, there are many challenges and opportunities that can be addressed in the future. First, the performance of categories that are harder to classify could be improved by integrating more advanced data augmentation techniques and using class-weighted loss functions to reduce misclassification. Additionally, future work could incorporate some class imbalance to represent real-world situations. Furthermore, testing the models on a purely real-world subset could provide deeper insights into generalisation capabilities outside controlled environments. Moreover, we could also integrate object detection to further improve it for practical applications by making it capable of identifying and classifying multiple waste objects within a single image. Finally, by expanding the dataset and using more efficient model compression techniques, we could be able to use high-performance models in edge devices or mobile applications, which would make it suitable for real-time and real-world recycling applications.

In conclusion, our work shows that deep learning models, such as CNNs, especially when combined with transfer learning, like EfficientNetB0 offers a powerful and scalable approach to waste classification. These models hold great potential for transforming recycling infrastructures and supporting more sustainable waste management systems. As environmental concerns grow and recycling systems strive for greater efficiency, such intelligent classification systems will be pivotal.

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